HW7

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Problem 1

$$\frac{\partial u}{\partial t} + \frac{\partial u^2}{\partial x} + \frac{\partial (x+y)u}{\partial y} = \nu\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial x \partial y}\right) + au^2$$

$$0 \le x \le L_x \qquad 0 \le y \le L_y$$

(a) FDE for interior points

Domain discretized as such: $x_i = (i-1)\Delta x$, $i = 1, 2, 3 \cdots, IL$

$y_j = (j-1)\Delta y$, $j = 1, 2, 3 \cdots, JL$

Let $u^n_{i,j}$ represent value of u at (i,j) spatial point & n time step

i) Self-starting second-order implicit for time differencing:

↳ Crank-Nicolson scheme : $\frac{\partial u}{\partial t} \approx \frac{u^{n+1}_{i,j} - u^n_{i,j}}{\Delta t} = \frac{1}{2}$

(ii) Second-order central differencing for spatial derivatives:

$$\frac{\partial u^2}{\partial x} = \frac{(u^n_{i+1,j})^2 - (u^n_{i-1,j})^2}{2\Delta x}$$

$$\frac{\partial (x+y)u}{\partial y} = \frac{(x_i + y_{j+1})u^n_{i,j+1} - (x_i + y_{j-1})u^n_{i,j-1}}{2\Delta y}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{u^n_{i+1,j} - 2u^n_{i,j} + u^n_{i-1,j}}{(\Delta x)^2}$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{u^n_{i,j+1} - 2u^n_{i,j} + u^n_{i,j-1}}{(\Delta y)^2}$$

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{u^n_{i+1,j+1} - u^n_{i+1,j-1} - u^n_{i-1,j+1} + u^n_{i-1,j-1}}{4\Delta x \Delta y}$$

(iii) $\frac{\partial u}{\partial t} = \nu\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial x \partial y}\right) + au^2 - \frac{\partial u^2}{\partial x} - \frac{\partial (x+y)u}{\partial y}$

$$\frac{u^{n+1}_{i,j} - u^n_{i,j}}{\Delta t} = \left[\nu\left(D_x^2 + D_y^2 + D_x D_y\right)u_{i,j} + \underbrace{a(u_{i,j})^2}_{\text{source term}} - D_x(u_{i,j})^2 - D_y(x_i + y_j)u_{i,j}\right]^{n+1}$$

⇒ Linearizing and writing in operator form:

$$(u^2_{i,j})^{n+1} \approx (u^2_{ij})^n + 2u^n_{ij}\underbrace{(u^{n+1} - u^n)}_{\Delta u^{n+1}_{ij}} + \mathcal{O}(\Delta t^2)$$

$$u^{n+1}_{ij} = (u^{n+1} - u^n) + u^n = \Delta u^{n+1}_{ij} + u^n_{ij}$$

$$\therefore \frac{\Delta u^{n+1}_{ij}}{\Delta t} = \left[\nu(D_x^2 + D_y^2 + D_x D_y)\underbrace{u^{n+1}_{ij}}_{\Delta u^{n+1}_{ij} + u^n_{ij}} + a\left((u^2_{ij})^n + 2u^n_{ij}\Delta u^{n+1}_{ij}\right) - D_x\left((u^2_{ij})^n + 2u^n_{ij}\Delta u^{n+1}_{ij}\right) - D_y(x_i+y_j)u^{n+1}_{ij}\right]$$

In operator form:

$$\Delta u^{n+1} - \Delta t\left[\nu(D_x^2 + D_y^2 + D_x D_y)\Delta u^{n+1} - a(2u^n \Delta u^{n+1}) + D_x(2u^n \Delta u^{n+1}) + D_y(x+y)\Delta u^{n+1}\right] = \left[\nu(D_x^2 + D_y^2 + D_z^2)u^n + a(u^2)^n - D_x(u^2)^n - D_y(x+y)u^n\right]\Delta t$$

We can treat the cross-derivative term explicitly (while maintaining second-order accuracy)

↳ $D_x D_y \Delta u^{n+1} \approx D_x D_y(u^n)$ → this term is then moves to the RHS

⇒ $$1 - \Delta t\left[\nu(D_x^2 + D_y^2) - a(2u^n) + D_x(2u^n) + D_y(x+y)\right]\Delta u^{n+1} = \overbrace{\left[\nu(D_x^2 + D_y^2 + D_z^2)u^n + a(u^2)^n - D_x(u^2)^n - D_y(x+y)u^n - D_x D_y u^n\right]\Delta t}^{\text{RHS}}$$

$$1 - \Delta t\left[\nu(D_x^2 + D_y^2) - a(2u^n) + D_x(2u^n) + D_y(x+y)\right]\Delta u^{n+1} = \text{RHS}$$

Now we can factor approximately (2 spatial dimensions), split the x & y factors & the apply the split operators at grid points ij & replace spatial derivatives by difference operators (these source term is put in w/ x factor — although this is not the best method)

$$\underbrace{\left\{1 - \Delta t\left[\nu D_x^2 - a2u^n + D_x(2u^n)\right]\right\}\left\{1 + \Delta t\left(\nu D_y^2 + D_y(x+y)\right)\right\}}_{\Delta u^*}\Delta u^{n+1} = \text{RHS}$$

⇒ $$\Delta u^*_{i,j} + \Delta t D_x^2 \Delta u^*_{i,j} - \Delta t a 2[u^n \Delta u^*]_{i,j} + \Delta t D_x(2u^n \Delta u^*)_{i,j} = \text{RHS}_{i,j}$$

$$\Delta u^*_{i,j} + \Delta t\left(\frac{\Delta u^*_{i+1,j} - 2\Delta u^*_{i,j} + \Delta u^*_{i-1,j}}{\Delta x^2}\right) - \Delta t a 2[u^n \Delta u^*]_{i,j} + \Delta t\left(\frac{(2u^n \Delta u^*)_{i+1,j} - (2u^n \Delta u^*)_{i-1,j}}{2\Delta x}\right) = \text{RHS}_{i,j}$$

$$L_{i,j}\,\Delta u^{*}_{i-1,j} + M_{i,j}\,\Delta u^{*}_{i,j} + N_{i,j}\,\Delta u^{*}_{i+1,j} = RHS_{i,j} \longrightarrow \text{Eq } (1)$$

$\left(\frac{\Delta t}{\Delta x^2} - \frac{2u^{n}_{i-1,j}\Delta t}{2\Delta x}\right)$ $\left(1 - \frac{2\Delta t}{\Delta x^2} - \Delta t\, a\, 2u^{n}_{i,j}\right)$ $\left(\frac{\Delta t}{\Delta x^2} + \frac{\Delta t\, 2u^{n}_{i+1,j}}{2\Delta x}\right)$ $\quad j = 1, 2, 3 \ldots JL-1$

Therefore, our tridiagonal matrix for interior points:

$$\begin{bmatrix} M_{2,j} & N_{2,j} & & & \\ L_{3,j} & M_{3,j} & N_{3,j} & & \\ & \ddots & \ddots & \ddots & \\ & & L_{JL-2,j} & M_{JL-2,j} & \\ & & & & M_{JL-1,j} \end{bmatrix} \begin{bmatrix} \Delta u^{*}_{2,j} \\ \Delta u^{*}_{3,j} \\ \vdots \\ \Delta u^{*}_{JL-1,j} \end{bmatrix} = \begin{bmatrix} RHS_{2,j} \\ RHS_{3,j} \\ \vdots \\ RHS_{JL-1,j} \end{bmatrix}$$

↳ Boundary conditions need to be handled separately (above solution only for interior points)

Similarly,

$$\Delta u^{*} = \left\{1 + \Delta t\,(\nu D_y^2 + D_y(v+y))\right\}\Delta u^{n+1}$$

$$= \left\{1 + \Delta t\,\nu\left(\frac{\Delta u^{n+1}_{i,j+1} - 2\,\Delta u^{n+1}_{i,j} + \Delta u^{n+1}_{i,j-1}}{\Delta y^2}\right) + \Delta t\left(\frac{(v_{i,j+1}+y_j)\,\Delta u^{n+1}_{i,j+1} - (v_{i,j-1}+y_j)\,\Delta u^{n+1}_{i,j-1}}{2\Delta y}\right)\right.$$

$$\Delta u^{*} = L_{i,j}\,\Delta u^{n+1}_{i,j-1} + M_{i,j}\,\Delta u^{n+1}_{i,j} + N_{i,j}\,\Delta u^{n+1}_{i,j+1} \longrightarrow \text{Eq } (2)$$

$\left(\frac{\Delta t\,\nu}{\Delta y^2} - \frac{(v_{i,j}+y_j)\Delta t}{2\Delta y}\right)$ $\left(1 - \frac{2\Delta t\,\nu}{\Delta y^2}\right)$ $\left(\frac{\Delta t\,\nu}{\Delta y^2} + \frac{\Delta t\,(v_{i,j}+y_j)}{2\Delta y}\right)$

A similar tridiagonal matrix can be constructed for Eq (2) for $i = 2, 3 \ldots IL-1$ which are the interior points

$$\begin{bmatrix} M_{i,2} & N_{i,2} & & & \\ L_{i,3} & M_{i,3} & N_{i,3} & & \\ & \ddots & \ddots & \ddots & \\ & & L_{i,JL-1} & M_{i,JL-2} & N_{i,JL-2} \\ & & & & M_{i,JL-1} \end{bmatrix} \begin{bmatrix} \Delta u^{n+1}_{i,2} \\ \Delta u^{n+1}_{i,3} \\ \vdots \\ \Delta u^{n+1}_{i,JL-1} \end{bmatrix} = \begin{bmatrix} \Delta u^{*}_{i,2} \\ \Delta u^{*}_{i,3} \\ \vdots \\ \Delta u^{*}_{i,JL-1} \end{bmatrix}$$

(b) Solution algorithm

1) Using Eq. (1) to solve for $\Delta u^{*}_{i,j}$ from $j = 2, 3 \ldots JL-1$ via Thomas algorithm
2) Use Eq (2) to solve for $\Delta u^{n+1}_{i,j}$ (using constructed tridiagonal matrix) from $i = 2, 3, \ldots IL-1$ using Thomas algorithm
3) Calculate $u^{n+1}_{i,j} = u^{n}_{i,j} + \Delta u^{n+1}_{i,j}$

↳ Repeat above steps until convergence of duration of interest.

Problem 2

Compressible, inviscid flow; ideal gas; 2D converging-diverging nozzle

$$\frac{\partial U}{\partial t} + \frac{\partial F}{\partial x} + \frac{\partial G}{\partial y} = 0$$

(a) Describe the problem; given where the Mach number and the static temperature at the duct inlet are 0.2 and 300 K, respectively and the pressure at the duct exit is 1 atm.

The given governing equation is the continuity equation for inviscid flow, where variables U, F, G are related to the flow quantities such as mass/density, velocity, pressure and energy. The 2D converging-diverging nozzle has only one inlet (left boundary) and outlet (right boundary), with no flow through the top and bottom walls

The Mach number represents the ratio of the flow velocity to the speed of sound in the medium. Since the Mach number at the inlet is 0.2, the flow is subsonic at the inlet.

(b) Formulate the governing equations for the problem by using the Euler equations with appropriate boundary conditions at the walls and the inflow and outflow boundaries.

The Euler equations for compressible, inviscid flow are given by:

1) Continuity eq.: $\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} = 0$

2) Momentum eq.: $\frac{\partial(\rho u)}{\partial t} + \frac{\partial}{\partial x}(\rho u^2 + P) + \frac{\partial}{\partial y}(\rho u v) = 0$

$\frac{\partial(\rho v)}{\partial t} + \frac{\partial}{\partial x}(\rho u v) + \frac{\partial}{\partial y}(\rho v^2 + P) = 0$

Combining these equations yields the given expression

$\frac{\partial U}{\partial t} + \frac{\partial F}{\partial x} + \frac{\partial G}{\partial y} = 0$

$$\frac{\partial(\rho v)}{\partial t} + \frac{\partial}{\partial x}(\rho u v) + \frac{\partial}{\partial y}(\rho v^2 + P) = 0$$

3) Energy equation : $\frac{\partial e}{\partial t} + \frac{\partial}{\partial x}(u(e+P)) + \frac{\partial}{\partial y}(v(e+P)) = 0$

$$\frac{\partial U}{\partial t} + \frac{\partial F}{\partial x} + \frac{\partial G}{\partial y} = 0$$

where $P = (\gamma - 1)\left(e - \frac{u^2+v^2}{2\rho}\right)$

↓ is given by Bernoulli's principle

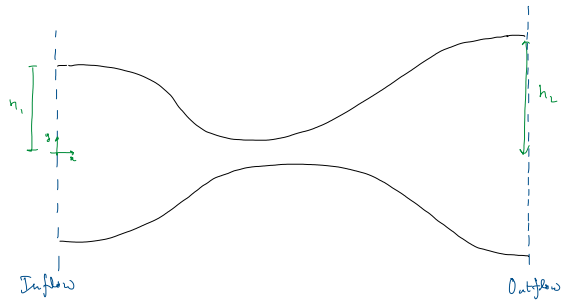
Boundary conditions :

@ walls : $u = v = 0$

@ outflow : $P = 1$ atm

@ inflow : $M = 0.2$, $T = 300$ K

For an ideal gas, {P,V,T} relationship is an additional equation that defines the variables.



(c) Generate a boundary-fitted grid for the 2-D nozzle geometry by using the two-boundary method, where grid lines should intersect boundary curves orthogonally.

$X_1(\xi,\tau) = x(\xi, \eta=0, \tau)$ $\quad X_2(\xi,\tau) = x(\xi, \eta=1, \tau)$

$Y_1(\xi,\tau) = y(\xi, \eta=0, \tau)$ $\quad Y_2(\xi,\tau) = y(\xi, \eta=1, \tau)$

$$\vec{r}_1 = X_1\vec{i} + Y_1\vec{j}$$

$$\vec{r}_L = X_2\vec{i} + Y_2\vec{j}$$

Hermite interpolation : $x(\xi,\eta,\tau) = X_1(\xi,\tau)h_1(\eta) + X_2(\xi,\tau)h_2(\eta) + \frac{\partial x(\xi,\eta=0,\tau)}{\partial \eta}h_3(\eta) + \frac{\partial x(\xi,\eta=1,\tau)}{\partial \eta}h_4(\eta)$

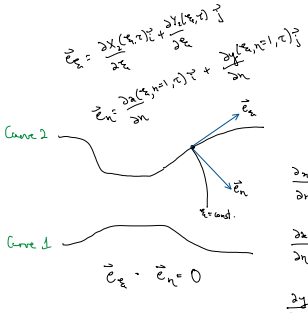
$y(\xi,\eta,\tau) = Y_1(\xi,\tau)h_1(\eta) + Y_2(\xi,\tau)h_2(\eta)$

$h_1(\eta) = 2\eta^3 - 3\eta^2 + 1$

$h_2(\eta) = -2\eta^3 + 3\eta^2$

$h_3(\eta) = \eta^3 - 2\eta^2 + \eta$

$h_4(\eta) = \eta^3 - \eta^2$



$$\frac{\partial x(\xi,\eta=1,\tau)}{\partial \eta} = -K_2(\xi,\tau)\frac{\partial Y_2(\xi,\tau)}{\partial \xi}$$

$$\frac{\partial x(\xi,\eta=0,\tau)}{\partial \eta} = +K_1(\xi,\tau)\frac{\partial Y_1(\xi,\tau)}{\partial \xi}$$

} K - functions

$$\frac{\partial y}{\partial \eta}(\xi,\eta=1,\tau) = \frac{\partial X_2}{\partial \xi} = \frac{C}{I_L}$$

$$\frac{\partial y}{\partial \eta}(\xi,\eta=0,\tau) = \frac{\partial X_1}{\partial \xi} = \frac{C}{I_L}$$

Parametrize the boundaries ; $x = \xi$ $\quad y = f(\lambda) = f(\xi)$

$$\vec{r} = \xi\,\vec{i} + f(\xi)\vec{j}$$

We are given :

Curves 1 $f_1(x)$:

$$\begin{cases} x \in [A,B] \;;\; Y_1 = (D-a_1) - a_1 \cos\left[\frac{\pi(x-A)}{(B-A)}\right] \\ x \in (B,C] \;;\; Y_1 = (D-a_2) + a_2 \cos\left[\frac{\pi(x-B)}{(C-B)}\right] \end{cases}$$

Curve 2 : $f_2(x)$

$$\begin{cases} x \in [A,B] \;;\; y_2 = (E+a_1) + a_1 \cos\left[\dfrac{\pi(x-A)}{(B-A)}\right] \\ x \in (B,C] \;;\; y_2 = (E+a_2) - a_2 \cos\left(\dfrac{\pi(x-B)}{(C-B)}\right) \end{cases}$$

$$y(\xi,\eta) = y_1\left(x(\xi)\right) + \eta\left(y_2(x(\xi)) - y_1(x(\xi))\right)$$

(d) Transform the governing equations and boundary conditions to boundary-fitted generalized coordinates.

$$(t,x,y) \rightarrow (\tau,\xi,\eta) \quad ; \quad \begin{aligned} \tau &= \tau(t,x,y) \\ \eta &= \eta(t,x,y) \\ \xi &= \xi(t,x,y) \end{aligned}$$

$$\frac{\partial U}{\partial t} + \frac{\partial F}{\partial x} + \frac{\partial G}{\partial y} = 0$$

$$\frac{\partial}{\partial t} = \underbrace{\frac{\partial \tau}{\partial t}}_{\tau_t}\frac{\partial}{\partial \tau} + \underbrace{\frac{\partial \eta}{\partial t}}_{\eta_t}\frac{\partial}{\partial \eta} + \underbrace{\frac{\partial \xi}{\partial t}}_{\xi_t}\frac{\partial}{\partial \xi}$$

$$\frac{\partial}{\partial x} = \frac{\partial \tau}{\partial x}\frac{\partial}{\partial \tau} + \frac{\partial \eta}{\partial x}\frac{\partial}{\partial \eta} + \frac{\partial \xi}{\partial x}\frac{\partial}{\partial \xi}$$

$$\frac{\partial}{\partial y} = \frac{\partial \tau}{\partial y}\frac{\partial}{\partial \tau} + \frac{\partial \eta}{\partial y}\frac{\partial}{\partial \eta} + \frac{\partial \xi}{\partial y}\frac{\partial}{\partial \xi}$$

$$\left(\tau_t\frac{\partial}{\partial \tau} + \eta_t\frac{\partial}{\partial \eta} + \xi_t\frac{\partial}{\partial \xi}\right)U + \left(\tau_x\frac{\partial}{\partial \tau} + \eta_x\frac{\partial}{\partial \eta} + \xi_x\frac{\partial}{\partial \xi}\right)F + \left(\tau_y\frac{\partial}{\partial \tau} + \eta_y\frac{\partial}{\partial \eta} + \xi_y\frac{\partial}{\partial \xi}\right)G = 0$$

↳ Transforming and converting to strong conservation form (multiply by J)

$$\Rightarrow \left[\frac{\partial}{\partial \tau}(J\tau_t U) - U\frac{\partial}{\partial \tau}(J\tau_t) + \frac{\partial}{\partial \eta}(J\eta_t U) - U\frac{\partial}{\partial \eta}(J\eta_t) + \frac{\partial}{\partial \xi}(J\xi_t U) - U\frac{\partial}{\partial \xi}(J\xi_t)\right]$$
$$+\left[\frac{\partial}{\partial \tau}(J\tau_x F) - F\frac{\partial}{\partial \tau}(J\tau_x) + \frac{\partial}{\partial \eta}(J\eta_x F) - F\frac{\partial}{\partial \eta}(J\eta_x) + \frac{\partial}{\partial \xi}(J\xi_x F) - F\frac{\partial}{\partial \xi}(J\xi_x)\right]$$
$$+\left[\frac{\partial}{\partial \tau}(\tau_y G J) - G\frac{\partial}{\partial \tau}(\tau_y J) + \frac{\partial}{\partial \eta}(\eta_y J G) - G\frac{\partial}{\partial \eta}(\eta_y J) + \frac{\partial}{\partial \xi}(\xi_y J G) - G\frac{\partial}{\partial \xi}(\xi_y J)\right] = 0$$

$$\Rightarrow \left[\frac{\partial}{\partial \tau}(J\tau_t U) + \frac{\partial}{\partial \eta}(J\eta_t U) + \frac{\partial}{\partial \xi}(J\xi_t U)\right] + \left[\frac{\partial}{\partial \tau}(J\tau_x F) + \frac{\partial}{\partial \eta}(J\eta_x F) + \frac{\partial}{\partial \xi}(J\xi_x F)\right] + \left[\frac{\partial}{\partial \tau}(\tau_y G J) + \frac{\partial}{\partial \eta}(\eta_y J G) + \frac{\partial}{\partial \xi}(\xi_y J G)\right]$$
$$-U\left[\frac{\partial}{\partial \tau}(J\tau_t) + \frac{\partial}{\partial \eta}(J\eta_t) + \frac{\partial}{\partial \xi}(J\xi_t)\right] - F\left[\frac{\partial}{\partial \tau}(J\tau_x) + \frac{\partial}{\partial \eta}(J\eta_x) + \frac{\partial}{\partial \xi}(J\xi_x)\right] - G\left[\frac{\partial}{\partial \tau}(J\tau_y) + \frac{\partial}{\partial \eta}(\eta_y J) + \frac{\partial}{\partial \xi}(\xi_y J)\right] = 0$$

If we assume $t = \tau$ & $\tau \neq f(x,y)$ & use geometric conservation laws
↳ Then our transformed governing equation (in strong conservative form) becomes:

$$\left[\frac{\partial}{\partial \tau}(J\tau_t U) + \cancel{\frac{\partial}{\partial \eta}(J\eta_t U)}^{\,0} + \cancel{\frac{\partial}{\partial \xi}(J\xi_t U)}^{\,0}\right] + \left[\cancel{\frac{\partial}{\partial \tau}(J\tau_x F)}^{\,0} + \frac{\partial}{\partial \eta}(J\eta_x F) + \frac{\partial}{\partial \xi}(J\xi_x F)\right] + \left[\cancel{\frac{\partial}{\partial \tau}(\tau_y G J)}^{\,0} + \frac{\partial}{\partial \eta}(\eta_y J G) + \frac{\partial}{\partial \xi}(\xi_y J G)\right]$$
$$-U\left[\frac{\partial}{\partial \tau}(J\tau_t) + \cancel{\frac{\partial}{\partial \eta}(J\eta_t)}^{\,0} + \cancel{\frac{\partial}{\partial \xi}(J\xi_t)}^{\,0}\right] - F\left[\frac{\partial}{\partial \tau}(J\tau_x) + \cancel{\frac{\partial}{\partial \eta}(J\eta_x)}^{\,0} + \cancel{\frac{\partial}{\partial \xi}(J\xi_x)}^{\,0}\right] - G\left[\frac{\partial}{\partial \tau}(J\tau_y) + \cancel{\frac{\partial}{\partial \eta}(\eta_y J)}^{\,0} + \cancel{\frac{\partial}{\partial \xi}(\xi_y J)}^{\,0}\right] = 0$$

$$\left[\frac{\partial}{\partial \tau}(J\tau_t U)\right] + \left[\frac{\partial}{\partial \tau}(J\tau_x F) + \frac{\partial}{\partial \eta}(J\eta_x F) + \frac{\partial}{\partial \xi}(J\xi_x F)\right] + \left[\frac{\partial}{\partial \tau}(\tau_y G J) + \frac{\partial}{\partial \eta}(\eta_y J G) + \frac{\partial}{\partial \xi}(\xi_y J G)\right] = 0$$

$$\therefore \boxed{\frac{\partial}{\partial \tau}(J\tau_t U) + \frac{\partial}{\partial \xi}(J\xi_x F + J\xi_y G) + \frac{\partial}{\partial \eta}(J\eta_x F + J\eta_y G) = 0}$$

$$\tilde{U} = \xi_x u + \xi_y v$$
$$\tilde{V} = \eta_x u + \eta_y v$$

Transforming the boundary conditions

inlet wall : $\xi = 0$ plane

outlet wall : $\xi = 1$ plane

bottom wall : $\eta = 0$ } Zero mass flux through walls $\quad \tilde{U}n_x - \tilde{V}\xi_x = 0$

top wall : $\eta = 1$ }

(e) Derive the FDE/FVEs for the governing equations by using the implicit Euler method with first-order upwinding using flux-vector splitting (any variation) that maintains the conservative property.

Flux-splitting scheme

$$\frac{\partial F}{\partial U} = \begin{bmatrix} \frac{\partial F_1}{\partial \rho} & \frac{\partial F_1}{\partial m} & \frac{\partial F_1}{\partial e} & \frac{\partial F_1}{\partial n} \\ \frac{\partial F_2}{\partial \rho} & \frac{\partial F_2}{\partial m} & \frac{\partial F_2}{\partial e} & \frac{\partial F_2}{\partial n} \\ \frac{\partial F_3}{\partial \rho} & \frac{\partial F_3}{\partial m} & \frac{\partial F_3}{\partial e} & \frac{\partial F_3}{\partial n} \\ \frac{\partial F_4}{\partial \rho} & \frac{\partial F_4}{\partial m} & \frac{\partial F_4}{\partial e} & \frac{\partial F_4}{\partial n} \end{bmatrix}$$

$$F = \begin{bmatrix} m \\ \frac{m^2}{\rho} + \gamma e - e - \frac{\gamma}{2\rho}(m^2+n^2) + \frac{(m^2+n^2)}{2\rho} \\ mn/\rho \\ \frac{e\rho}{\rho} + \frac{m}{\rho}\gamma e - \frac{\rho\gamma e}{\rho} - \frac{m}{2\rho^2}\gamma(m^2+n^2) + \frac{m}{2\rho^2}(m^2+n^2) \end{bmatrix}$$

$$\frac{\partial G}{\partial U} = \begin{bmatrix} \frac{\partial G_1}{\partial \rho} & \frac{\partial G_1}{\partial m} & \frac{\partial G_1}{\partial e} & \frac{\partial G_1}{\partial n} \\ \frac{\partial G_2}{\partial \rho} & \frac{\partial G_2}{\partial m} & \frac{\partial G_2}{\partial e} & \frac{\partial G_2}{\partial n} \\ \frac{\partial G_3}{\partial \rho} & \frac{\partial G_3}{\partial m} & \frac{\partial G_3}{\partial e} & \frac{\partial G_3}{\partial n} \\ \frac{\partial G_4}{\partial \rho} & \frac{\partial G_4}{\partial m} & \frac{\partial G_4}{\partial e} & \frac{\partial G_4}{\partial n} \end{bmatrix}$$

$$G = \begin{bmatrix} n \\ mn/\rho \\ n^2/\rho + \gamma e - e - \gamma\left(\frac{m^2+n^2}{2\rho}\right) + \frac{(m^2+n^2)}{2\rho} \\ \frac{e\rho}{\rho} + \frac{n}{\rho}\gamma e - \frac{\rho\gamma e}{\rho} - \frac{\gamma}{\rho}\left(\frac{m^2+n^2}{2\rho}\right) + \frac{n}{2\rho^2}(m^2+n^2) \end{bmatrix}$$

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{m^2}{\rho^2} + \frac{\gamma}{2}(m^2+n^2) - \frac{(m^2+n^2)}{\rho^2} & \frac{2m}{\rho} - \frac{\gamma m}{\rho} + \frac{m}{\rho} & (\gamma-1) & \frac{\gamma n}{\rho} + \frac{n}{\rho} \\ -\frac{mn}{\rho^2} & \frac{n}{\rho} & 0 & \frac{m}{\rho} \\ \left(-\frac{\gamma e}{\rho^2} + \frac{(n\gamma - m)(m^2+n^2)}{\rho^2}\right) & \left(-\frac{\gamma e}{\rho} - \frac{3\gamma m^2}{2\rho^2} + \frac{3m^2}{2\rho}\right) & \frac{m\gamma}{\rho} & \frac{\gamma mn}{\rho^2} + \frac{mn}{\rho^2} \end{bmatrix}$$

$$B = \begin{bmatrix} 0 & 0 & 0 & 1 \\ -\frac{mn}{\rho^2} & \frac{n}{\rho} & 0 & \frac{m}{\rho} \\ \left(-\frac{n^2}{\rho^2} + \frac{\gamma(m^2+n^2)}{2\rho^2}\right) & \left(-\frac{\gamma n}{\rho} + \frac{n}{\rho}\right) & (\gamma-1) & \left(-\frac{\gamma n}{\rho} + \frac{n}{\rho}\right) \\ -\frac{n\gamma e}{\rho^2} + \frac{2(\gamma n(m^2+n^2))}{2\rho^3} - \frac{n(m^2+n^2)}{\rho^3} & \left(\frac{\gamma n m}{\rho^2} + \frac{2mn}{\rho^2}\right) & & \left(\frac{\gamma e}{\rho} + \frac{3\gamma n^2}{2\rho^2} + \frac{3n^2}{2\rho^2}\right) \end{bmatrix}$$

$F = AU = (A^+ + A^-)U = F^+ + F^-$

positive eigenvalues, negative eigenvalues

Flux splitting in non-transformed domain:

$$\therefore \frac{\partial U}{\partial t} + \frac{\partial}{\partial x}(F^+ + F^-) + \frac{\partial}{\partial y}(G^+ + G^-) = 0$$

$$\frac{\partial U}{\partial t} + \frac{\partial F^+}{\partial x} + \frac{\partial F^-}{\partial x} + \frac{\partial G^+}{\partial y} + \frac{\partial G^-}{\partial y} = 0$$

$$\frac{U_{ij}^{n+1} - U_{ij}^n}{\Delta t} + \frac{F_{ij}^+ - F_{i-1,j}^+}{\Delta x} + \frac{\hat{F}_{i+1,j}^- - F_{ij}^-}{\Delta x} + \frac{G_{ij}^+ - G_{ij-1}^+}{\Delta y} + \frac{G_{ij+1}^- - G_{ij}^-}{\Delta y} = 0$$

For euler implicit, F^+, F^-, G^+ & G^- are all at time level $n+1$

↳ Similar logic applied on the transformed equations

⇒ Central differencing outside derivatives

$$\frac{\partial}{\partial \tau}(J\tau_t U) + \frac{\partial}{\partial \xi}(J\xi_x F + J\xi_y G) + \frac{\partial}{\partial \eta}(J\eta_x F + J\eta_y G) = 0$$

$$\frac{(J\tau_t U)_{i+\frac{1}{2},j}^n - (J\tau_t U)_{i-\frac{1}{2},j}^n}{\Delta\tau} \qquad \frac{(J\xi_x F + J\xi_y G)_{i+\frac{1}{2},j}^n - (J\xi_x F + J\xi_y G)_{i-\frac{1}{2},j}^n}{\Delta\xi}$$

↳ we can apply this on the transformed equation

(f) Develop an algorithm to generate solutions to the FDE/FVEs that uses approximate factorization.

Algorithm:

1) Discretize domain: $\Delta\eta$, $\Delta\xi$, Δt
   - ↳ choose IL, JL
   - ↳ set $\Delta\eta = \Delta\xi = 1$

2) Solve for Y_1, Y_2, X_1, X_2, $\frac{dY_1}{d\xi}$, $\frac{dY_2}{d\xi}$ & h coefficients

3) Euler implicit time-differencing + linearization

$$\frac{U_{ij}^{n+1} - U_{ij}^n}{\Delta t} = -\left(\frac{\partial F}{\partial x} + \frac{\partial G}{\partial y}\right)_{ij}^{n+1}$$

$$[I + \Delta t(D_x A + D_y B)]\Delta U_{ij}^{n+1} = RHS_{ij}$$

$D_x A$, $D_y B$: from above transformed equations

$$RHS = -\Delta t\left(\frac{\partial F}{\partial x} + \frac{\partial G}{\partial y}\right)_{ij}^n$$

4) Approximate factorization in spatial direction

$$\left\{I + \Delta t\left[D_x(A^+ + A^-) + D_y(B^+ + B^-)\right]\right\}\Delta U_{ij}^{n+1} = RHS_{ij}$$

$$\left[I + \Delta t(L+K)\right]\Delta U_{ij}^{n+1} = RHS_{ij}$$

$$L = S_x^b A^+ + S_y^b B^+ \quad ; \quad K = S_x^f A^- + S_y^f B^-$$

S^b & S^f are backward & forward difference operators

4) The equations are solved by splitting the operator yields (similar to problem 1)

(g) Write a computer program for the algorithm.

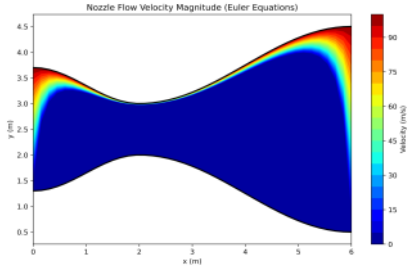
I have tried to program the above algorithm in my code (used LLM for debugging and help) - find attached.

(h) Obtain the steady-state solution

For steady state solutions, LHS can be 1st order accurate but RHS is second order accurate

The steady state solution is when there is not time dependence i.e. the solution is time invariant and the governing equations become just

$$\frac{\partial F}{\partial x} + \frac{\partial G}{\partial y} = 0$$



(i) Discuss accuracy of results with a grid sensitivity study.

There seems to be a bug in the code that I was not able to resolve due to time constraint so my analysis is only limited to the results I obtained. Here are some expected outcomes/analysis based on the physics of converging-diverging nozzle: we expect that the maximum velocity should reach asymptotic value with grid refinement and finer grids should show decreasing differences in the parameters.

Further refining the grid did not change the contour plot solution. Therefore, the observed solution convergence across grid refinements indicates that discretization errors from mesh resolution were negligible. Hence, the remaining numerical errors most likely are from the accuracy of finite difference formulation (assuming no bugs in the code).